REAL-TIME QUALITY MONITORING IN PRODUCTION USING SCADA SYSTEMS AT HLL

Project Report

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in

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DEPARTMENT OF MECHANICAL ENGINEERING

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DECLARATION

We undersigned hereby declare that the project report (“Real-Time Quality Monitoring in Production using SCADA Systems at HLL”) , submitted for partial fulfillment of the requirements for the award of degree of Master of Technology of the APJ Abdul Kalam Technological University, Kerala is a bonafide work done by me under supervision of Prof. Jijith P K . This submission represents my ideas in our own words and where ideas or words of others have been included, we have adequately and accurately cited and referenced the original sources. We also declare that we have adhered to ethics of academic honesty and integrity and have not misrepresented or fabricated any data or idea or fact or source in this submission. We understand that any violation of the above will be a cause for disciplinary action by the institute and/or the University and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been obtained. This report has not been previously formed the basis for the award of any degree, diploma or similar title of any other University.

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Certificate

This is to certify that the project report entitled **" Real-Time Quality Monitoring in Production using SCADA Systems at HLL"** is a bonafide record of the project work carried out by **Anchitha Balamurali (TVE22IE013), Meenakshi P(TVE22IE044), Nithin S Kumar(TVE22IE051), Sanusha Sajeev (TVE22IE061)** under my guidance and supervision, submitted to APJ Abdul Kalam Technological University in partial fulfillment of the requirements for the Bachelor of Technology in Industrial Engineering.

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ABBREVIATIONS

Abbreviation	Full Form
HLL	HLL Lifecare Limited
SCADA	Supervisory Control and Data Acquisition
IoT	Internet of Things
IIoT	Industrial Internet of Things
PLC	Programmable Logic Controller
ETD	Electronic Testing Department
HPQA	Half Product Quality Assurance
FPQA	Full Product Quality Assurance
QA	Quality Assurance
CPS	Cyber-Physical System
HMI	Human–Machine Interface
RTU	Remote Terminal Unit
MTU	Master Terminal Unit
SPC	Statistical Process Control
OEE	Overall Equipment Effectiveness
SOP	Standard Operating Procedure
RCA	Root Cause Analysis
EIA	Environmental Impact Assessment
QC	Quality Control
TPM	Total Productive Maintenance
SPM	Standard Performance Measure
TQM	Total Quality Management

ABSTRACT

Quality control in condom manufacturing is of paramount importance due to its direct impact on public health, safety, and product reliability. At HLL Lifecare Ltd., the Electronic Testing Department (ETD) is responsible for detecting defects in condoms using electrical testing methods. However, the existing quality control system is predominantly manual and lacks real-time integration, resulting in several operational challenges. These include false rejection of good-quality condoms, inconsistent classification accuracy due to environmental variations such as humidity and sensor misalignment, delayed detection of process deviations, and limited visibility into machine and operator performance. These issues contribute to increased material wastage, higher production costs, and reduced overall efficiency.

This project aims to address these challenges by developing and implementing a SCADA (Supervisory Control and Data Acquisition) based real-time monitoring system for the ETD process. The proposed system integrates sensors and programmable logic controllers (PLCs) to continuously monitor critical parameters such as test voltage, test current, environmental conditions, and operator efficiency. The SCADA system provides a centralized platform for real-time data acquisition, visualization, alarm generation, and historical data storage, enabling better process control and decision-making.

The methodology involves detailed process study, data collection, identification of critical parameters, and the design of a SCADA-based monitoring architecture. Statistical tools such as control charts (P-charts) are used to analyze process performance before and after implementation. The results demonstrate a significant improvement in process stability, with all data points falling within control limits after SCADA implementation. There is a noticeable reduction in variability, improved consistency in product classification, and a substantial decrease in false rejection rates.

Furthermore, the system enhances operational efficiency by enabling early detection of abnormalities through alarms and alerts, reducing downtime and facilitating timely corrective actions. It also improves data accuracy and ensures centralized data management, providing valuable insights for continuous improvement.

In conclusion, the implementation of a SCADA-based real-time monitoring system transforms the quality control process from a manual and reactive approach to an automated and proactive system. This not only improves product quality and reduces waste but also enhances productivity and cost-effectiveness. The developed framework serves as a scalable and adaptable solution that can be extended to other quality-sensitive manufacturing sectors, aligning with Industry 4.0 principles and promoting sustainable manufacturing practices.
Keywords:SCADA, Real-time Monitoring, Quality Control, ETD, P-chart, Industrial Automation, Process Stability, Manufacturing Efficiency

Chapter 1

INTRODUCTION

In the realm of healthcare manufacturing, product quality is not just a metric—it is a mandate. At HLL Lifecare Ltd., a leading public sector enterprise specializing in the production of condoms and other medical products, quality control plays a pivotal role in ensuring user safety, regulatory compliance, and brand integrity. However, the current quality assurance process relies heavily on manual inspection and limited automation, which often leads to the false rejection of good-quality products, increased material waste, and operational inefficiencies. This project aims to address these challenges by designing and implementing a real-time quality monitoring system using Supervisory Control and Data Acquisition (SCADA) integrated with Programmable Logic Controllers (PLCs). SCADA systems are widely recognized for their ability to provide centralized monitoring, data logging, and intelligent control across industrial processes. By leveraging these capabilities, the proposed system will continuously track critical parameters such as test voltage, test current, and operator efficiency, enabling early detection of deviations and more accurate classification of product quality. The integration of SCADA into HLL's production line is expected to significantly reduce false rejections, minimize material waste, and enhance overall productivity. Moreover, the project aligns with Industry 4.0 principles, promoting digital transformation and data-driven decision-making in traditionally manual sectors. The developed framework will not only benefit HLL but also serve as a scalable model for other quality-sensitive manufacturing environments such as pharmaceuticals, food processing, and medical devices. By combining industrial engineering methodologies with advanced automation technologies, this initiative represents a forward-looking approach to sustainable manufacturing and continuous improvement. It reflects a commitment to innovation, operational excellence, and public health impact.

This report is organized into seven chapters to systematically present the study on real-time quality monitoring using SCADA systems at HLL Lifecare Ltd. Chapter 1 introduces the project background, objectives, and significance, along with a detailed company profile and an overview of the Electronic Testing Department (ETD). Chapter 2 presents a comprehensive literature review, highlighting previous research on SCADA, PLC integration, and Industry 4.0 technologies in manufacturing. Chapter 3 defines the problem statement, outlines the motivation behind the study, and specifies the objectives and scope. Chapter 4 details the methodology adopted, including process study, critical parameter identification, SCADA-based monitoring design, and strategies for quality improvement and scalability. Chapter 5 analyzes the causes of defects in quality checking, such as improper covering, vulcanization issues, and sensor errors. Chapter 6 discusses the expected results, including improved quality visibility, reduced false rejections, and enhanced productivity. Finally, Chapter 7 concludes the report by summarizing key findings and outlining the potential impact of the proposed system, followed by the references section.

1.1 COMPANY PROFILE

HLL LIFECARE LIMITTED

The motivation for this project stems from the critical need to ensure the safety, reliability, and sustainability of pipeline networks. With an increasing global demand for water, oil, gas, and other resources transported through pipelines, infrastructure failures due to cracks can have significant economic, environmental, and social impacts. Efficient, accurate, and timely detection of cracks in pipelines can prevent potential failures, reduce maintenance costs, and increase public safety. As machine learning and robotics technologies continue to advance, there is a compelling opportunity to leverage these tools for autonomous, non-invasive pipeline inspections that can detect and predict faults before they escalate. This project aims to address these needs by developing an in-pipe Remotely Operated Vehicle (ROV) equipped with machine learning-based crack detection, offering a scalable solution to meet the challenges faced by current inspection methods.

1.2 ELECTRONIC TESTING DEPARTMENT (ETD)

The Electronic Testing Department (ETD) plays a crucial role in ensuring the reliability and safety of condoms manufactured at HLL Lifecare Ltd. The main objective of this department is to identify any defective products—particularly those with pinholes, thin spots, or electrical leaks—that may compromise product quality. This stage is vital because condoms are medical-grade products where safety and quality cannot be compromised. ETD ensures that every single condom undergoes 100% electrical testing to detect even the smallest defects, thereby upholding HLL's commitment to international quality standards.

Working Principle

The ETD process is based on the principle of electrical conductivity. Each condom is carefully mounted on a stainless-steel mould, which serves as an electrode. When an electrical voltage is applied, the system measures the current that passes through the condom. If there are no holes or defects, the condom acts as an insulator, and no current flows. However, if there is a pinhole or a thin area, current passes through the latex, signaling a defect. The testing equipment automatically classifies the condoms—those without any electrical leakage are marked as good, while those with current flow are identified as defective and separated. This automated system enables accurate, quick, and reliable testing for every piece produced.

Process Flow in ETD

After the vulcanization process in the primary production section, the condoms are transferred to the Electronic Testing Department for inspection. The process begins with operators mounting each condom on stainless steel moulds to ensure full and proper coverage. These moulds then pass through the electronic testing machine, which subjects them to a fixed voltage and current. The machine continuously monitors for any electrical leaks that indicate defects. Based on the readings, the machine automatically sorts the condoms into good and bad trays. After sorting, the counts of good and defective condoms are recorded and verified against machine data to ensure accuracy. This systematic process guarantees that only defect-free products proceed to the packing section.

Selectivity Checking Process

To ensure the accuracy and sensitivity of the ETD machines, a selectivity checking process is performed regularly. This test is conducted every two hours and at the start of each shift. During this process, a sample of twenty condoms from the "good" category is taken, and tiny holes are manually created in them using a hypodermic needle of 0.25 mm (31 gauge). These artificially punctured condoms are mixed with defect-free ones and tested again using the ETD machine. The machine must correctly identify all defective condoms for it to pass the selectivity test. If any punctured condoms go undetected, recalibration or maintenance of the sensors is carried out. This ensures that the ETD equipment maintains 100% accuracy and reliability in defect detection throughout production.

QUALITY ASSURANCE

Quality Assurance (QA) in the Electronic Testing Department (ETD) is an integral part of the production process at HLL Lifecare Ltd., ensuring that every product meets stringent safety and performance standards. Since condoms are medical-grade products directly linked to public health, the quality assurance procedures in ETD are designed to detect even the smallest imperfections that may compromise product integrity. The QA system functions through a combination of automated electronic inspection, statistical sampling, and manual verification, ensuring complete reliability and accuracy in defect identification. By implementing structured testing, calibration, and documentation practices, the QA team in ETD maintains consistency in production, minimizes false rejections, and ensures that only defect-free products proceed to the packing stage. This comprehensive quality assurance framework reflects HLL's commitment to precision, safety, and global manufacturing standards. Quality Assurance in the Electronic Testing Department (ETD) at HLL Lifecare Ltd. is carried out in three systematic levels to ensure that every product meets the highest quality and safety standards before packaging and dispatch.

- Half product Quality Assurance (HPQA)
- ETD online process
- Full product Quality Assurance

Each level focuses on a different stage of production and collectively ensures that defects are identified and eliminated effectively, minimizing the chances of defective products reaching the end user.

Half Product Quality Assurance

The first stage of quality assurance, known as Half Product Quality Assurance (HPQA), is conducted immediately after the vulcanization process and before the condoms enter the electronic testing section. In this stage, visual inspection plays a key role. A random sample of 40 condoms is selected from each lot and examined for any surface irregularities such as bubbles, uneven thickness, or deformities. These visual checks help to identify manufacturing inconsistencies that may affect the electrical testing process or product performance. By identifying and addressing such issues at an early stage, HPQA ensures that only defect-free half products proceed to the ETD for final electrical testing. This preventive step helps maintain the accuracy of the electronic inspection process and reduces unnecessary rejections later.

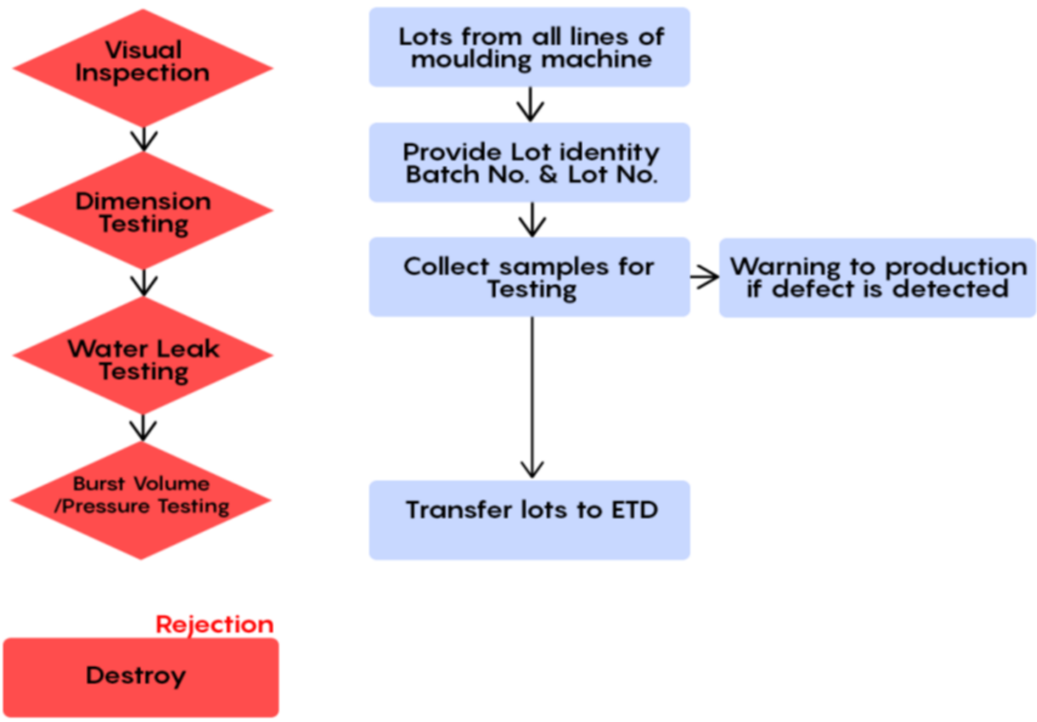

Figure 1.1: Half Product Quality Assurance Testing Process(HPQA)

ETD online process

The second stage, known as the ETD Online Process, is conducted during the electronic testing operation itself. This step serves as an in-process validation of the ETD machines to ensure that they are functioning accurately and classifying products correctly. After about 90 of the testing in ETD is completed, additional samples are taken for verification—typically 24 samples for domestic batches and 30 for export batches. These samples are rechecked under controlled conditions to confirm the accuracy of the machine's classifications. Any discrepancies between machine readings and actual results are analyzed to identify calibration issues or sensor misalignments. This stage is vital to maintaining continuous process control, ensuring that the ETD machines are reliable and performing within the required sensitivity range.

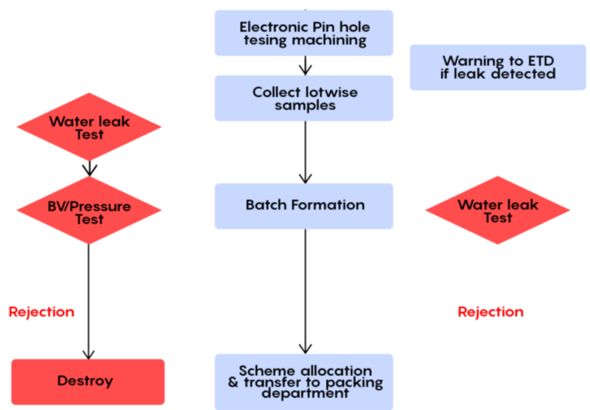

Figure 1.2: ETD Online QA Testing

Finished Product Quality Assurance

The final stage of quality assurance is Full Product Quality Assurance (FPQA), which involves testing and verifying the quality of the fully finished condoms after electronic testing and before pac king. In this stage, the performance of the ETD system and the accuracy of classification are thoroughly evaluated. A minimum of twenty condoms are selected from each unit, and their good and bad counts are compared with actual performance results. Calibration checks are also performed by taking ten condoms from the good tray, unrolling them over moulds in the opposite testing unit, and verifying that all are correctly identified as good. Any mismatch or inconsistency detected in this phase triggers corrective actions such as recalibration, re-inspection, or process adjustment. FPQA acts as the final verification stage to ensure that all condoms approved by ETD meet international safety standards and customer requirements.

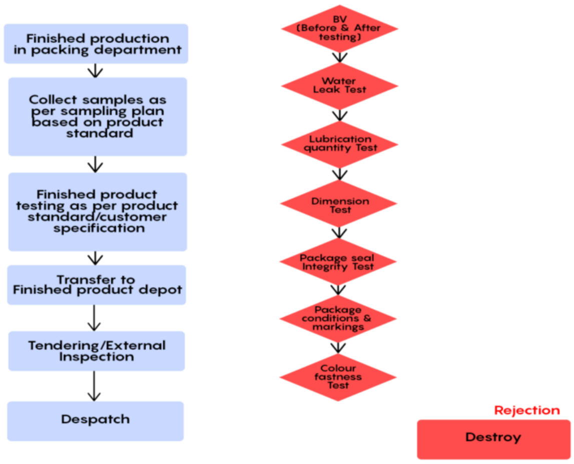

Figure 1.3: Finished Product Quality Assurance

Performance checking

* Checking is done once every shift after the selectivity check and within two hours from beginning of shift

* Cover minimum twenty condoms in one unit, continuously note down good and bad condoms

* Collect the condoms falling into good and bad trays and count them

* Record count good , actual good as well as count bad and actual bad. check for any discrep-

ancies

Good condom Calibration

Once in every shift

* Take ten pieces of good from good tray and unroll it safely over the moulds in opposite unit .

Check whether all the above covered good condoms identified as good by testing unit.

Chapter 2

LITERATURE REVIEW

The evolution of industrial automation over the past few decades has been deeply influenced by the emergence of Supervisory Control and Data Acquisition (SCADA) systems, Programmable Logic Controllers (PLCs), and more recently, Industry 4.0 technologies. The integration of these systems has led to significant improvements in process monitoring, control, and optimization across a wide range of industries. Researchers have explored various aspects of these technologies—from their architectural design and implementation to their role in enhancing security, efficiency, and product quality. The following review summarizes and analyzes key contributions from previous studies that collectively illustrate how SCADA and Industry 4.0 technologies are transforming manufacturing and industrial systems into intelligent, data-driven environments.

Alcaraz and Lopez (2018) conducted one of the most comprehensive reviews on the architecture and security of SCADA systems, presenting valuable insights into how these systems have evolved from isolated, proprietary networks into open and interconnected structures. Their study, published in Computers Security, identifies the major components of SCADA—such as the Master Terminal Unit (MTU), Remote Terminal Units (RTUs), communication networks, and Human-Machine Interfaces (HMIs)—and explains their roles in industrial process monitoring. The authors note that early SCADA systems operated in physically isolated environments, which inherently protected them from external threats. However, as industries transitioned to networked and Internet-based SCADA systems for improved communication and data sharing, new vulnerabilities emerged. Cybersecurity has become a critical concern because modern SCADA systems are now exposed to cyberattacks, malware, and unauthorized access attempts. Alcaraz and Lopez argue that traditional security mechanisms are inadequate in addressing these new challenges and propose a multi-layered security framework that includes network segmentation, encryption, authentication, and continuous anomaly detection. The paper also connects the development of SCADA security with the broader movement toward Industry 4.0, suggesting that future industrial environments will rely heavily on secure, resilient, and adaptive control systems. Their work establishes the foundation for understanding the importance of security in automated manufacturing, especially as industries adopt interconnected smart systems.[1]

Building upon the foundational understanding of SCADA, Kumar and Yadav (2020) explored its practical implementation in industrial automation and process control. Their study, published in the International Journal of Engineering Research Technology, focuses on how SCADA can enhance the operational efficiency and productivity of industrial plants. Through case studies and real-world implementations, the authors describe how SCADA systems integrate field devices such as sensors, actuators, and PLCs to enable centralized control and monitoring. They emphasize that SCADA provides not only real-time data visualization but also the capability to log and analyze historical data, allowing industries to identify inefficiencies and predict equipment failures. The authors also note that the flexibility and scalability of SCADA make it suitable for diverse industrial environments, ranging from power generation and water treatment to manufacturing and chemical processing. One of the major contributions of this paper is its detailed discussion of how HMI platforms and communication protocols such as Modbus and Ethernet facilitate seamless connectivity among devices.

The study concludes that implementing SCADA systems significantly reduces operational downtime, improves decision-making, and enhances overall process reliability. This research effectively demonstrates how automation technologies are helping industries transition from manual, reactive management systems to proactive and data-driven control environments.[2]
A further advancement in SCADA applications was discussed by Singh and Gupta (2021), who focused on the integration of PLC and SCADA systems for real-time monitoring in manufacturing industries. Their work, published in the International Journal of Advanced Research in Electrical, Electronics and Instrumentation Engineering, highlights the synergy between PLCs—which execute control logic at the machine level—and SCADA systems, which provide supervisory control and centralized visualization. The authors explain that this integration allows for continuous monitoring of key process variables, automated alarm generation, and real-time feedback control. The study presents examples from manufacturing units where PLC-SCADA integration has led to improved production efficiency, reduced response time to faults, and enhanced transparency in operations. The research also underscores the importance of interoperability between different hardware and software platforms, achieved through standardized communication protocols. Singh and Gupta point out that in an Industry 4.0 context, such integration acts as a stepping stone toward fully digitized smart factories where machines, systems, and humans communicate seamlessly to achieve self-optimizing production processes. Their findings reaffirm that integrating automation layers—ranging from machine control to data supervision—creates an intelligent ecosystem capable of maintaining consistent product quality and operational excellence.[3]
Similarly, Shukla and Patel (2019) examined real-time data acquisition and monitoring using SCADA in process industries, as presented in the Journal of Automation and Control Engineering. Their study emphasizes the importance of continuous data collection and visualization for ensuring efficient process performance. By integrating SCADA with sensors and distributed control systems, they demonstrate how industries can achieve continuous

tracking of parameters such as pressure, temperature, and flow rate, thereby ensuring that processes remain within desired operational limits. The authors also discuss the implementation of alarm systems and graphical dashboards that allow plant operators to instantly identify and correct anomalies. One of the significant outcomes highlighted in their research is the reduction of process variability through early detection of deviations. The paper also identifies technical challenges such as network latency and data synchronization that can limit the real-time capabilities of SCADA. Despite these limitations, the authors conclude that SCADA remains an indispensable technology for achieving reliability, safety, and operational continuity in complex industrial environments. The insights from their study demonstrate how effective real-time monitoring forms the foundation for higher-level applications such as predictive maintenance and automated quality control, which are key goals in the transition toward Industry 4.0.[4]

The evolution from conventional automation to intelligent manufacturing has been conceptually captured by Lee, Bagheri, and Kao (2015) in their pioneering paper titled A Cyber-Physical Systems Architecture for Industry 4.0-Based Manufacturing Systems, published in Manufacturing Letters. The authors introduce a five-level cyber-physical systems (CPS) architecture that outlines how raw data can be transformed into intelligent decisions within manufacturing environments. Their proposed model consists of the smart connection, data-to-information conversion, cyber, cognition, and configuration levels, each representing a step toward self-adaptive, autonomous production systems. Lee and his colleagues emphasize that while traditional SCADA systems primarily handle monitoring and control, CPS frameworks extend these capabilities through integration with cloud computing, data analytics, and machine learning. This integration enables predictive maintenance, intelligent scheduling, and system optimization in real-time. The authors further describe how CPS facilitates the creation of digital twins—virtual representations of physical assets that allow simulation, monitoring, and optimization of manufacturing operations. By aligning cyber and physical processes, manufacturers can achieve a high level of flexibility, efficiency, and quality consistency. The study also highlights key enablers of Industry 4.0, including interoperability, real-time data exchange, and decentralized decision-making. Overall, their work provides a theoretical and architectural foundation that connects traditional automation technologies such as SCADA with modern digital manufacturing paradigms.[5]

While the architectural and implementation aspects of SCADA and Industry 4.0 are essential, recent research has placed greater emphasis on how these systems can enhance product quality and maintenance efficiency through data-driven approaches. Zhang and Li (2020), in their paper Quality Monitoring and Predictive Maintenance Using Data-Driven Approaches in Smart Manufacturing published in Procedia Manufacturing, focus on leveraging data analytics to predict equipment failures and improve product quality. Their study demonstrates how integrating data from sensors, SCADA, and other industrial control systems allows manufacturers to apply machine learning algorithms for fault prediction and process optimization. The authors discuss various techniques, such as regression models and neural networks, for analyzing sensor data and detecting anomalies before they result in process deviations or equipment breakdowns. They highlight the importance of data preprocessing, feature selection, and real-time model updates to ensure accuracy and adaptability in dynamic manufacturing environments. Through case studies, Zhang and Li show that predictive maintenance significantly reduces downtime, extends equipment life, and enhances overall equipment effectiveness (OEE). The paper concludes that the combination of SCADA data and advanced analytics creates a new paradigm for intelligent quality management, where systems not only monitor operations but also anticipate and prevent failures.[6]

Building upon this data-driven approach, Karthikeyan and Rajkumar (2022) investigated how SCADA systems can be used to reduce product rejection rates through real-time quality monitoring. Their research, published in the International Journal of Industrial Engineering Technology, emphasizes the role of SCADA in ensuring continuous quality control by integrating it with automated feedback mechanisms. The authors report that by collecting real-time data on critical process parameters and establishing threshold-based alerts, industries can identify quality deviations immediately and take corrective actions before defective products are produced. They further discuss how integrating SCADA with statistical process control (SPC) tools can help analyze trends and identify root causes of variation. The study presents case examples from discrete manufacturing sectors where implementing SCADA-based monitoring reduced rework, scrap, and rejection rates significantly. The authors conclude that such systems are instrumental in achieving the concept of zero-defect manufacturing, which is a central goal of Industry 4.0. Their findings highlight that when SCADA systems are enhanced with data analytics and closed-loop feedback, they transition from being merely monitoring tools to becoming active components in quality assurance and process improvement.[7]

Collectively, the reviewed studies present a clear picture of how SCADA systems and Industry 4.0 technologies are converging to create intelligent, interconnected, and efficient manufacturing systems. Alcaraz and Lopez (2018) laid the groundwork by addressing the crucial issue of security in increasingly connected control systems. Kumar and Yadav (2020) demonstrated how SCADA implementation improves automation and operational reliability, while Singh and Gupta (2021) expanded on its integration with PLCs to enable real-time decision-making. Shukla and Patel (2019) contributed to understanding the significance of continuous data acquisition and visualization in maintaining process consistency. Lee et al. (2015) bridged traditional automation with the future vision of Industry 4.0 through their cyber-physical systems model, while Zhang and Li (2020) and Karthikeyan and Rajkumar (2022) highlighted how data analytics and SCADA integration can drive predictive maintenance and quality improvement.

Despite these advancements, the literature reveals that several challenges remain in achieving full integration of SCADA with advanced data-driven systems. One major challenge is ensuring the interoperability of legacy systems with modern digital architectures. Many existing SCADA installations still operate on proprietary protocols and hardware, making seamless integration with Industry 4.0 platforms difficult. Another challenge involves cybersecurity, as increased connectivity exposes systems to potential cyber threats. Furthermore, while predictive maintenance and quality monitoring have been widely researched, their large-scale industrial implementation often requires high computational resources, standardized data formats, and skilled personnel capable of managing and interpreting complex datasets.

The reviewed studies collectively affirm that the convergence of SCADA, CPS, and data analytics is central to the realization of smart manufacturing and zero-defect production. These technologies are transforming industrial operations from reactive to proactive systems, where decisions are guided by data, and every process is continuously optimized. This literature provides the conceptual and technical basis for further exploration into developing real-time monitoring solutions that enhance product quality and reduce false rejection in industrial environments. Such systems align closely with the vision of Industry 4.0 and demonstrate how integrating automation, security, and analytics can lead to sustainable manufacturing excellence.

2.1 RESEARCH GAP

Despite the proven effectiveness of SCADA and PLC systems in various industrial applications, their use in real-time quality monitoring within condom manufacturing remains underexplored. Existing literature focuses primarily on process automation, fault detection, and safety enhancement in sectors like pharmaceuticals, glass manufacturing, and chemical processing. However, the following gaps persist:

• Limited Application of SCADA in Quality Control: Most implementations emphasize process control rather than product-level quality classification, especially in healthcare manufacturing.

• Lack of Real-Time Quality Classification: Current systems do not integrate machine and operator data to classify product quality dynamically, leading to false rejections and inefficiencies.

• Insufficient Focus on Safety, Security, and Regulatory Compliance: Few studies address how SCADA systems can be tailored to meet the stringent safety and compliance standards required in condom production.

This project seeks to bridge these gaps by developing a SCADA-based framework specifically designed for real-time quality monitoring and classification in condom manufacturing at HLL.

Chapter 3

PROBLEM DEFINITION

The current quality control process at HLL Lifecare Ltd. is predominantly manual and lacks real-time integration, resulting in:

• False rejection of good-quality condoms, leading to material waste and increased production costs.

• Inconsistent classification accuracy, due to environmental factors like humidity and sensor misalignment.

• Limited visibility into operator performance and machine-level deviations, which hinders timely corrective actions.

To address these issues, the project proposes the development of a SCADA-based real-time monitoring system that:

• Continuously tracks critical parameters such as test voltage, test current, and operator efficiency. • Accurately classifies product quality to minimize false rejections.

• Enhances production efficiency and provides a scalable model for other quality-sensitive manufacturing sectors.

3.1 MOTIVATION

Quality assurance in condom manufacturing is not merely a technical requirement—it is a public health imperative. At HLL Lifecare Ltd., ensuring that every product meets stringent safety and performance standards is critical. However, the current reliance on manual inspection and limited automation often results in the false rejection of good-quality condoms, leading to increased material waste, higher production costs, and reduced operational efficiency.

While SCADA and PLC systems have demonstrated success in other industries—such as pharmaceuticals, chemical processing, and glass manufacturing—their application in condom production remains minimal. This gap presents a unique opportunity to harness real-time monitoring technologies to enhance quality control in a sector where precision and reliability are non-negotiable.

By implementing a real-time integrated monitoring system, this project aims to:

• Reduce waste and false rejections through accurate classification.

• Improve production efficiency by enabling early detection of deviations.

• Establish a scalable model for other quality-sensitive manufacturing sectors.

This initiative is driven by the dual goals of technological innovation and sustainable manufacturing, aligning with Industry 4.0 principles and HLL's commitment to excellence.

3.2 OBJECTIVES

The primary objectives of this project are:

• To analyze critical process parameters—such as test voltage, test current, and operator efficiency—that directly influence product quality in condom manufacturing.

• To evaluate the effectiveness of SCADA and PLC integration for real-time monitoring, with the aim of minimizing false rejection rates and material waste.

• To develop a SCADA-based real-time quality monitoring system that integrates both machine-level and operator-level data to improve classification accuracy and decision-making.

• To enhance production efficiency and quality visibility, thereby contributing to a more reliable and cost-effective manufacturing process.

• To create a scalable and adaptable framework that can be extended to other healthcare and quality-sensitive industries in the future.

3.3 SCOPE

This project focuses on the design and implementation of a SCADA-based real-time quality monitoring system within the condom manufacturing process at HLL Lifecare Ltd. The scope includes:

• Integration of critical parameters such as test voltage, test current, and operator efficiency to enable accurate product quality classification.

• Reduction of false rejections of good-quality condoms, thereby minimizing material waste and improving cost-effectiveness.

• Enhancement of production efficiency through early detection of deviations and real-time feedback mechanisms.

• Development of a scalable framework that can be adapted to other quality-sensitive manufacturing sectors such as pharmaceuticals, food processing, and medical devices.

• Support for continuous improvement by enabling data-driven decision-making and standardizing quality control procedures.

Chapter 4

METHODOLOGY

The methodology adopted for this project follows a systematic industrial engineering approach, combining process analysis, automation design, and continuous improvement strategies. The goal is to develop a SCADA-based real-time quality monitoring system tailored to the condom manufacturing process at HLL Lifecare Ltd.

4.1 PROCESS STUDY DATA COLLECTION

The Process Study and Data Collection phase forms the fundamental basis for the system design and improvement framework in the project. This step involves a comprehensive understanding of the existing condom manufacturing process at HLL Lifecare Ltd., with the objective of identifying quality-critical operations, recurring faults, and parameters that influence product performance. By carefully studying the production workflow, data sources, and defect patterns, this stage ensures that the proposed SCADA-based real-time quality monitoring system is designed with precise insight into the current manufacturing challenges.

The first stage of this process involves mapping the entire production workflow, starting from latex compounding and mould dipping to vulcanization, electronic testing, and final packaging. Each of these stages contributes significantly to the overall product quality. Through detailed observation and interaction with process engineers, the study identifies how variations in process conditions—such as temperature, curing time, or operator handling—affect the final output. Mapping the workflow not only provides a visual representation of material and information flow but also helps in identifying interdependencies between departments, such as the influence of the vulcanization section on the performance of the Electronic Testing Department (ETD).

The next step focuses on identifying quality-sensitive stages within the process. Among all the production units, the Electronic Testing Department (ETD) emerges as the most critical area since it directly determines the acceptance or rejection of the final product. This department often faces challenges like false rejections, which can result from environmental factors such as high humidity, unstable temperature, or improper sensor calibration. These false rejections lead to unnecessary wastage and reduced productivity. Therefore, this phase emphasizes detailed monitoring of the ETD environment and testing setup to recognize parameters that significantly influence defect identification and classification accuracy.

Following this, extensive efforts are made to collect historical data from production records, operator logs, and quality assurance reports. The data primarily includes process parameters such as ambient temperature, relative humidity, test voltage, test current, and operator performance. These datasets are gathered from both automated logs and manual registers maintained by the ETD and QA departments. The collected information helps in understanding the normal operating ranges and deviations that correlate with variations in rejection rates. In addition, machine maintenance and calibration records are reviewed to assess the impact of equipment performance on test accuracy.

The final step of this phase involves analyzing rejection patterns using statistical tools and data visualization methods. Techniques such as trend analysis, control charts, and correlation studies are applied to evaluate recurring issues and anomalies. By studying historical data over several production lots, the analysis helps to identify patterns such as higher rejection rates during specific shifts, environmental conditions, or machine configurations. This insight is essential for pinpointing the root causes of false rejections and understanding how process fluctuations affect quality classification.

Overall, the Process Study and Data Collection phase provides a solid empirical foundation for designing the SCADA-based monitoring system. It ensures that the proposed solution is not built on assumptions but on factual, data-driven evidence gathered from the real production environment at HLL Lifecare Ltd. This systematic approach bridges the gap between existing process inefficiencies and the technological framework required for achieving accurate, reliable, and real-time quality monitoring in the Electronic Testing Department

4.2 CRITICAL PARAMETER IDENTIFICATION

a) Statistical Correlation Analysis

Statistical correlation analysis is used to understand the relationship between various process parameters and product quality outcomes. By examining historical production data, such as test voltage, current, temperature, and humidity, alongside product rejection records, it is possible to quantify the influence of each variable on quality performance. Techniques like correlation coefficients, regression analysis, or multivariate analysis help identify which parameters have the strongest impact on false rejections. For example, in the Electronic Testing Department, fluctuations in test voltage or environmental humidity may correlate highly with defective readings, making them key focus areas. This analysis enables the team to distinguish between critical and non-critical parameters, ensuring that monitoring and improvement efforts are directed where they are most needed.

b)Threshold Setting

Threshold setting involves defining the acceptable operating ranges for each critical parameter to maintain product quality within desired limits. These ranges are determined based on historical performance data, industry standards, and best practice guidelines. Clear thresholds ensure that any deviation outside the acceptable limits can be quickly detected and corrected, preventing quality issues before they affect the final product. For instance, if test voltage is found to cause defects when below 4.8V or above 5.2V, the process is controlled within this band. Setting precise thresholds not only safeguards quality but also provides objective criteria for operators and automated systems to manage production consistently.

c) Prioritization

Prioritization focuses on identifying which critical parameters require continuous monitoring and which can be managed periodically. Parameters that are highly sensitive to environmental changes, operator handling, or equipment variability are given the highest priority, as even small deviations can result in significant quality issues. Less sensitive parameters may be checked at regular intervals without compromising product reliability. By prioritizing monitoring efforts, resources are used efficiently, and the risk of false rejections or defects is minimized. For example, in a condom manufacturing line, humidity and test voltage may be monitored in real-time due to their direct impact on electronic testing results, while other parameters like machine calibration may be reviewed less frequently.

4.3 SCADA-BASED MONITORING DESIGN

The SCADA (Supervisory Control and Data Acquisition) phase represents the core of automation in modern manufacturing systems, providing centralized control and real-time monitoring of critical process parameters. Its design and implementation ensure that production can be continuously monitored, deviations can be promptly addressed, and historical data is systematically recorded for analysis and process optimization.

a) System Architecture

The first step involves designing a robust SCADA framework that interfaces seamlessly with the existing PLCs (Programmable Logic Controllers) on the shop floor. The system architecture defines how sensors, PLCs, and the SCADA software communicate, ensuring reliable data acquisition and control. This framework allows supervisors to monitor and control multiple machines from a central location, reducing manual intervention and enabling quicker response to any abnormalities. The architecture also incorporates redundancy and fail-safe mechanisms to maintain system reliability in case of hardware or network failures.

b) Dashboard Development

Real-time dashboards are created as the operator interface for the SCADA system. These dashboards display key process metrics, such as test voltage, current, environmental conditions, and operator activity, in a user-friendly format. By providing instant visibility into machine performance and production status, dashboards enable supervisors and quality engineers to make informed decisions quickly. Customizable dashboards allow critical parameters to be highlighted and grouped by machine, production line, or department, providing a comprehensive view of operations at a glance.

c) Visualization Tools

SCADA systems leverage advanced visualization tools to present process data in an intuitive manner. Trend charts, graphical indicators, and color-coded gauges are used to track parameter fluctuations over time, making it easier to detect patterns, anomalies, or gradual drifts that could lead to quality issues. Visualization facilitates predictive maintenance and process optimization by allowing engineers to analyze historical trends and correlate parameter behavior with product quality outcomes.

d) Alarm and Alert Systems

Threshold-based alarms are configured within the SCADA system to alert operators whenever critical parameters exceed predefined limits. These alarms can be visual, auditory, or even sent via mobile notifications to ensure immediate awareness of potential issues. By providing timely alerts for deviations that could result in false rejections, the system minimizes downtime and reduces the risk of defective products reaching the next stage of production. Operators can respond to alarms proactively, adjusting settings or performing corrective actions to maintain optimal process conditions.

e) Data Logging

Continuous data acquisition and storage form the backbone of SCADA-based monitoring. Every parameter, machine status, and operator intervention is recorded in a secure database, enabling historical analysis, traceability, and regulatory compliance. Logged data helps identify recurring issues, supports process improvement initiatives, and provides evidence for audits and quality certifications. Over time, this data becomes a valuable asset for predictive analytics, process optimization, and decision-making.

4.4 QUALITY ANALYSIS IMPROVEMENT STRATEGY

Once the SCADA-based monitoring system is operational, industrial engineering techniques are employed to systematically analyze process performance and enhance quality control. These tools help in identifying the most significant sources of defects, monitoring process stability, and implementing corrective actions to improve overall product quality.

a) Pareto Analysis

Pareto analysis is used to identify the most frequent causes of defects and false rejections by applying the "80/20 rule," which states that roughly 80% of problems are caused by 20%

of the root causes. Data collected from the SCADA system, such as defect type, machine, or process parameter, is analyzed to prioritize issues based on frequency and impact. This allows the quality team to focus resources on the critical few factors that contribute most to production inefficiencies or false rejections. For example, if sensor misalignment accounts for the majority of false rejections, corrective measures can be prioritized in that area.

b) Control Charts

Control charts are employed to monitor the stability and consistency of the production process in real time. By plotting key process parameters such as test voltage, current, or environmental conditions against upper and lower control limits, deviations or trends that indicate potential quality issues can be detected early. When a parameter exceeds these control limits, it signals an out-of-control condition, prompting immediate investigation. This proactive monitoring prevents the accumulation of defects and ensures that the process remains within desired quality standards.

c) Root Cause Analysis

Root cause analysis (RCA) is conducted to investigate the underlying reasons for defects and process deviations. Using data from SCADA logs, operators and engineers systematically examine potential causes, such as sensor misalignment, improper vulcanization, equipment wear, or environmental interference. Techniques like the “5 Whys” or Fishbone (Ishikawa) diagrams are applied to trace problems to their source rather than addressing only the symptoms. Identifying root causes ensures that corrective actions are effective and sustainable.

d) Targeted Corrective Actions

Insights gained from Pareto analysis, control charts, and root cause analysis guide the implementation of targeted corrective actions. Adjustments may include recalibrating sensors, refining process parameters, enhancing operator training, or modifying equipment settings. By addressing the most impactful causes of defects, these interventions improve classification accuracy, reduce false rejections, and enhance overall product quality. Continuous monitoring and iterative improvements create a feedback loop that sustains process optimization over time.

4.5 EVALUATION CONTINUOUS IMPROVEMENT

After the SCADA system is fully implemented, it is essential to assess its effectiveness in improving process quality and efficiency, and to make continuous refinements based on observed performance and operator experience. This step ensures that the system delivers sustained benefits and remains aligned with production requirements.

a) Performance Comparison

The first aspect of evaluation involves comparing key performance metrics before and after SCADA implementation. Historical production data, such as false rejection rates, material waste, machine downtime, and overall process efficiency, are analyzed to quantify improvements. By contrasting pre-implementation baseline data with post-implementation results, the impact of the SCADA system on product quality and operational efficiency can be objectively measured. For example, a reduction in false rejections or a decrease in wasted materials directly indicates the system's effectiveness in controlling critical parameters and supporting informed decision-making.

b) System Refinement

Based on the performance analysis, the SCADA system is refined to enhance its functionality and responsiveness. This may include adjusting alarm thresholds to better detect deviations, modifying reporting formats to highlight the most relevant metrics, or reorganizing dashboard layouts for easier visualization of critical information. Continuous refinement ensures that the system remains adaptive to process changes, prevents false alerts, and maintains accurate real-time monitoring. These incremental improvements help optimize operator response times and sustain high-quality production standards.

c) Operator Feedback

Feedback from shop-floor personnel is an integral component of system optimization. Operators who interact with the SCADA system daily provide practical insights on usability, alert effectiveness, and dashboard navigation. Incorporating their suggestions enhances the system's ergonomics and responsiveness, making it more intuitive and easier to use. By combining operator experience with quantitative performance metrics, the SCADA system becomes a more effective tool for real-time monitoring, decision-making, and continuous quality improvement.

4.6 SCALABILITY STANDARDIZATION

The final step of the SCADA-based monitoring project focuses on ensuring that the developed system is not only effective in the initial implementation but also scalable, sustainable, and replicable across other production lines or facilities. This step is critical for institutionalizing best practices, supporting long-term quality improvements, and promoting responsible manufacturing.

a) Framework Development

A scalable and modular framework is created to facilitate the deployment of the SCADA-based monitoring system across multiple production lines or even different facilities. This framework standardizes system architecture, communication protocols, and sensor integration methods, enabling consistent implementation without extensive customization. By designing the system with scalability in mind, HLL ensures that lessons learned from one production line can be efficiently applied elsewhere, saving time, cost, and effort while maintaining process reliability and quality standards.

b) Documentation

Comprehensive documentation is prepared to support consistent and accurate system deployment. This includes Standard Operating Procedures (SOPs) for daily operations, detailed training manuals for operators and engineers, technical specifications for hardware and software components, and guidelines for system maintenance. Thorough documentation ensures that personnel at all levels can understand, operate, and maintain the system effectively, reducing reliance on individual expertise and promoting standardization across the organization.

c) Compliance Assurance

The SCADA system is aligned with relevant regulatory standards, industry best practices, and data integrity protocols to ensure long-term compliance and reliability. This includes adherence to quality certifications, electronic data management regulations, and cybersecurity measures. Ensuring compliance not only safeguards the organization against legal or regulatory issues but also enhances the credibility and reliability of the monitoring system in the eyes of stakeholders, auditors, and regulatory authorities.

d) Sustainability Focus

Beyond technical replication, the project emphasizes sustainability and continuous improvement. The system encourages responsible manufacturing practices, such as minimizing material waste, reducing false rejections, and optimizing energy use. By embedding continuous monitoring, real-time feedback, and iterative improvements into daily operations, HLL can maintain high-quality standards while promoting environmentally and economically sustainable production. This approach also provides a blueprint for similar industries seeking to implement automated quality control systems while supporting broader corporate social responsibility goals.

Chapter 5

CAUSES OF DEFECTS IN QUALITY CHECKING

Quality checking in condom manufacturing, particularly in electronic testing, is highly sensitive to variations in process, environment, and equipment. False rejections of good-quality condoms not only increase material waste but also disrupt production efficiency and compromise overall quality assurance. Understanding the root causes of these defects is essential for implementing effective corrective measures.

5.1 IMPROPER COVERING

Improper covering occurs when condoms are not fully extended to the grooves on the steel mandrel during the dipping process. Complete coverage ensures uniform contact between the condom surface and the electrodes of the electronic testing device. When condoms are partially covering the mandrel, electrodes may not touch the entire surface, resulting in incomplete testing. Consequently, the device may register abnormal readings and classify good condoms as defective. This problem highlights the importance of precise mandrel placement, consistent dipping procedures, and operator training. Automation or vision-based verification systems can also ensure proper extension before testing, reducing the likelihood of false rejections.

5.2 IMPROPER VULCANIZATION

Vulcanization is a critical process that imparts elasticity and strength to latex. Insufficient vulcanization—due to short curing time, inadequate temperature, or improper chemical conditions—leaves condoms wet or tacky. Wet surfaces are more conductive, which causes the electronic tester to detect higher current or voltage readings than normal. This leads to good condoms being incorrectly flagged as defective. Ensuring strict adherence to vulcanization parameters, regular monitoring of curing conditions, and sample checks are vital to prevent this type of defect and maintain consistent product quality.

5.3 HIGH AMBIENT HUMIDITY

Excess humidity in the Electronic Testing Device (ETD) area can cause condensation on the surface of condoms. This condensation creates unintended conductive pathways between the electrodes, which can trigger false positives during leak detection tests. High ambient humidity is particularly challenging in tropical climates or during seasonal variations. To mitigate this, controlled environmental conditions—using dehumidifiers, air-conditioning, or proper ventilation—are essential. Continuous monitoring of temperature and humidity ensures stable testing conditions and improves reliability of defect detection.

5.4 SENSOR POSITIONING LOSSAGE ERRORS

Sensors in electronic testing devices must be precisely positioned and maintained to accurately classify condoms. Misaligned, loose, or malfunctioning sensors may fail to detect defects or may incorrectly classify products. Faulty sensors may either allow defective condoms to pass or reject good condoms, thereby reducing process efficiency and increasing waste. Regular sensor calibration, preventive maintenance, and real-time monitoring of sensor performance are critical to avoid these errors. Additionally, automated alerts for sensor misalignment can minimize downtime and improve testing reliability.

5.5 MATERIAL WEAKNESS

Inconsistent latex thickness or weak spots can occur due to variations in dipping, latex formulation, or curing. Thin areas conduct electricity more readily and are more susceptible to damage during testing. Even if the majority of the condom is defect-free, these weak regions can lead the electronic tester to incorrectly identify the product as defective. Standardizing latex formulations, refining dipping techniques, and incorporating automated thickness measurement or inspection systems help reduce the incidence of material weakness and prevent false rejections.

5.6 LACK OF DOWNSTREAM FEEDBACK

Many electronic testing systems do not include secondary verification or feedback mechanisms after initial classification. Once a condom passes or fails a sensor test, there is no mechanism to confirm or correct the decision. This lack of downstream feedback means that false positives remain uncorrected and actual defects may go undetected, compromising overall quality assurance. Integrating feedback loops, such as automated re-tests for borderline products or cross-verification sensors, can enhance system reliability and reduce the rate of false rejections.

The causes of defects in electronic quality checking are multifaceted, arising from process inconsistencies, environmental factors, and equipment limitations. Each factor—improper covering, vulcanization issues, high humidity, sensor errors, material weaknesses, and lack of feedback—can independently or collectively contribute to false rejections of good-quality condoms. Addressing these issues requires a combination of process standardization, environmental control, equipment maintenance, and system feedback mechanisms. Implementing these measures enhances testing accuracy, reduces waste, and improves overall production efficiency, ensuring that only high-quality products reach the market.

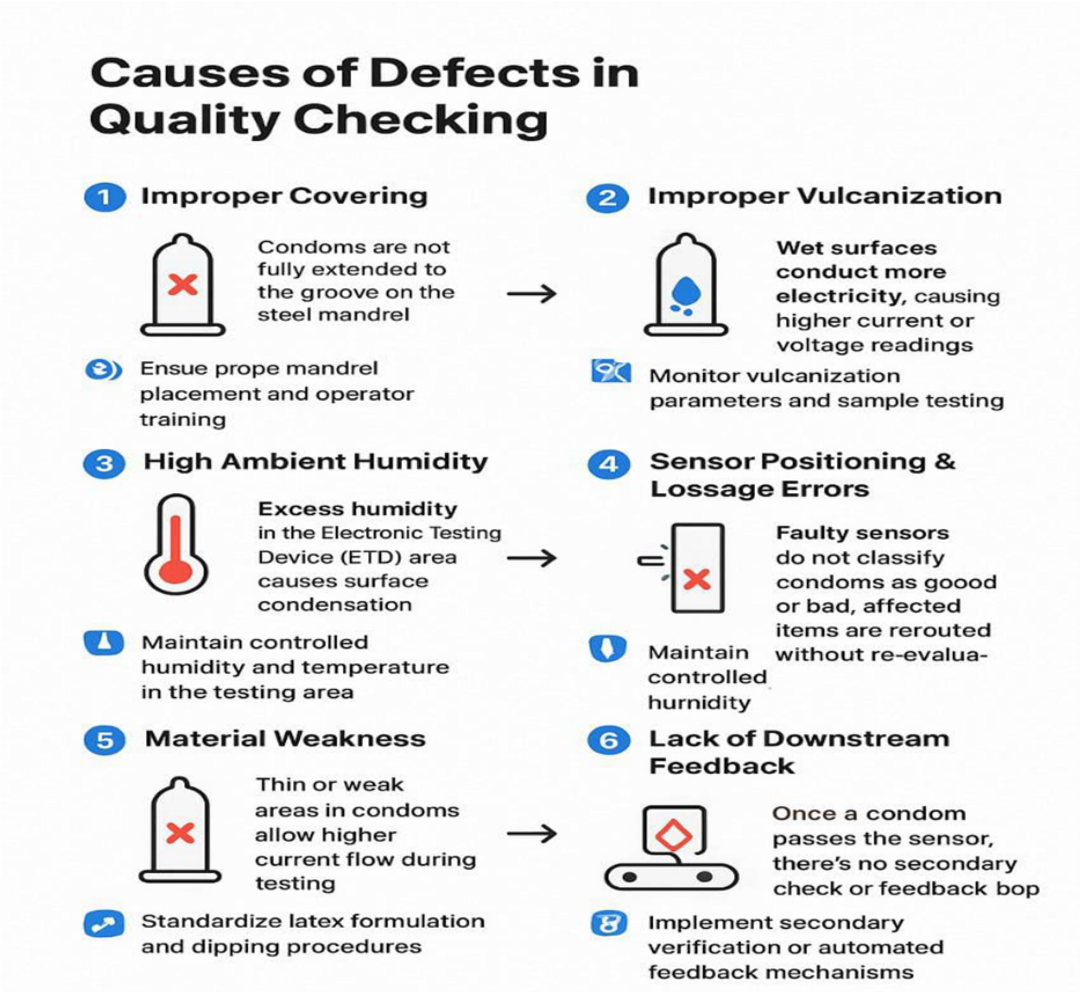

Figure 5.1: causes of defect in quality checking.

5.7 ROOT CAUSE ANALYSIS OF POOR EFFICIENCY IN ETD DATA CAPTURING

The problem of poor efficiency in data capturing in the Electronic Testing Department (ETD) can be attributed to multiple factors related to man, machine, material, and method. From the human perspective, lack of proper training in process control, delayed performance testing, and operator fatigue contribute to inconsistencies in monitoring and decision-making.

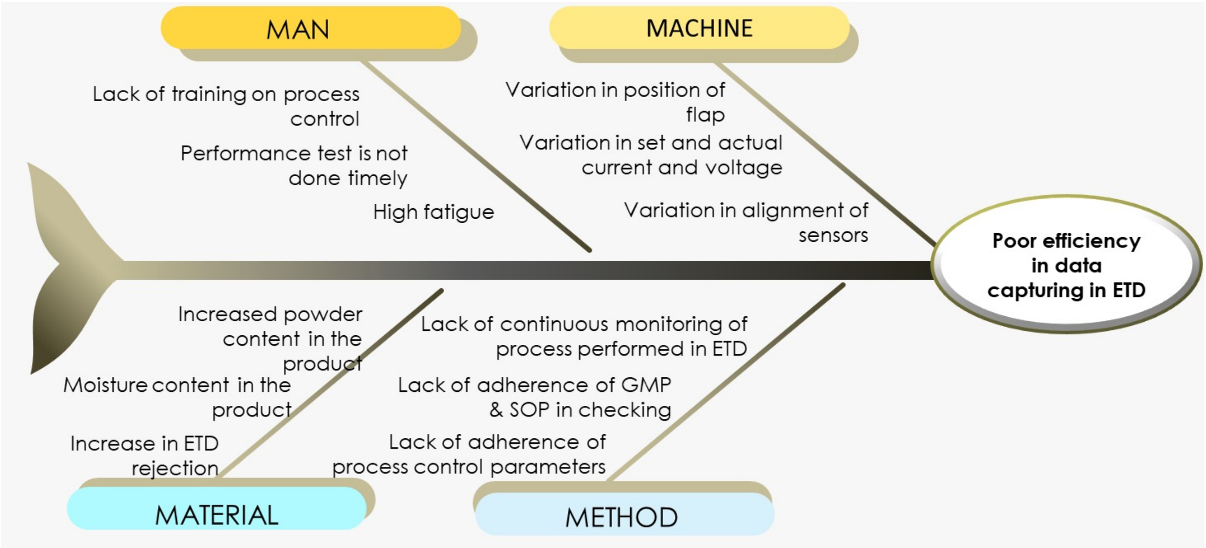

Figure 5.2: Fish bone diagram

Machine-related issues such as variation in flap positioning, differences between set and actual current and voltage, and misalignment of sensors lead to inaccurate readings and classification errors. Material factors, including high moisture content, increased powder content, and rising rejection rates, further affect the reliability of testing outcomes. Additionally, methodological shortcomings such as lack of continuous monitoring, poor adherence to GMP and SOP standards, and inadequate process control practices result in inefficient data handling. Together, these factors create significant challenges in achieving accurate, consistent, and efficient quality control in the ETD.

Chapter 6

Performance Analysis Before SCADA

The current performance of the Electronic Testing Department (ETD) indicates several inefficiencies due to the lack of real-time monitoring and automation. In-depth analysis capability is limited to around 50% as data is not systematically collected or analyzed. Real-time data monitoring is only about 40%, which means most decisions are based on delayed or manually recorded information. Similarly, decision-making efficiency is also around 40%, as operators do not have immediate access to accurate data.

Centralized data collection is very low, approximately 20%, highlighting the absence of an integrated system to store and analyze production data. One of the major concerns is the delay in identifying issues, which is as high as 80%, resulting in late corrective actions and increased defects. The rejection rate is around 10%, which includes both actual defects and false rejections of good-quality products. Additionally, overall productivity is only about 60%, indicating significant room for improvement.

These limitations clearly show that the existing system is inefficient and reactive in nature. Therefore, implementing a real-time monitoring system such as SCADA can address these issues by improving visibility, reducing delays, and enhancing decision-making.

In depth analysis	~50%
Real time data monitoring	~40%
Decision making	~40%
Centralized data collection	~20%
Delay in identifying issues	~80%
Rejection rate	~10%
Productivity	~60%

6.1 P-CHART BEFORE SCADA IMPLEMENTATION

The P-chart represents the proportion of defective products across different batches before the implementation of SCADA. From the chart, it is evident that the process is not stable and is statistically out of control.

Several data points exceed the Upper Control Limit (UCL), which is approximately 0.070, indicating that certain batches have an unusually high number of defects. Additionally, a few points fall below the Lower Control Limit (LCL), around 0.019, suggesting inconsistencies in inspection or measurement errors.

There is also a wide variation in defect proportions across batches, ranging from approximately 0.008 to 0.088. This high variability indicates poor process consistency and lack of control over critical parameters. Furthermore, there is no clear trend or pattern observed in the chart, meaning the variation appears random and is due to common causes rather than a specific assignable cause.

The average defect proportion is around 4.4%, which is relatively high for a quality-sensitive product like condoms. This clearly indicates the need for immediate corrective

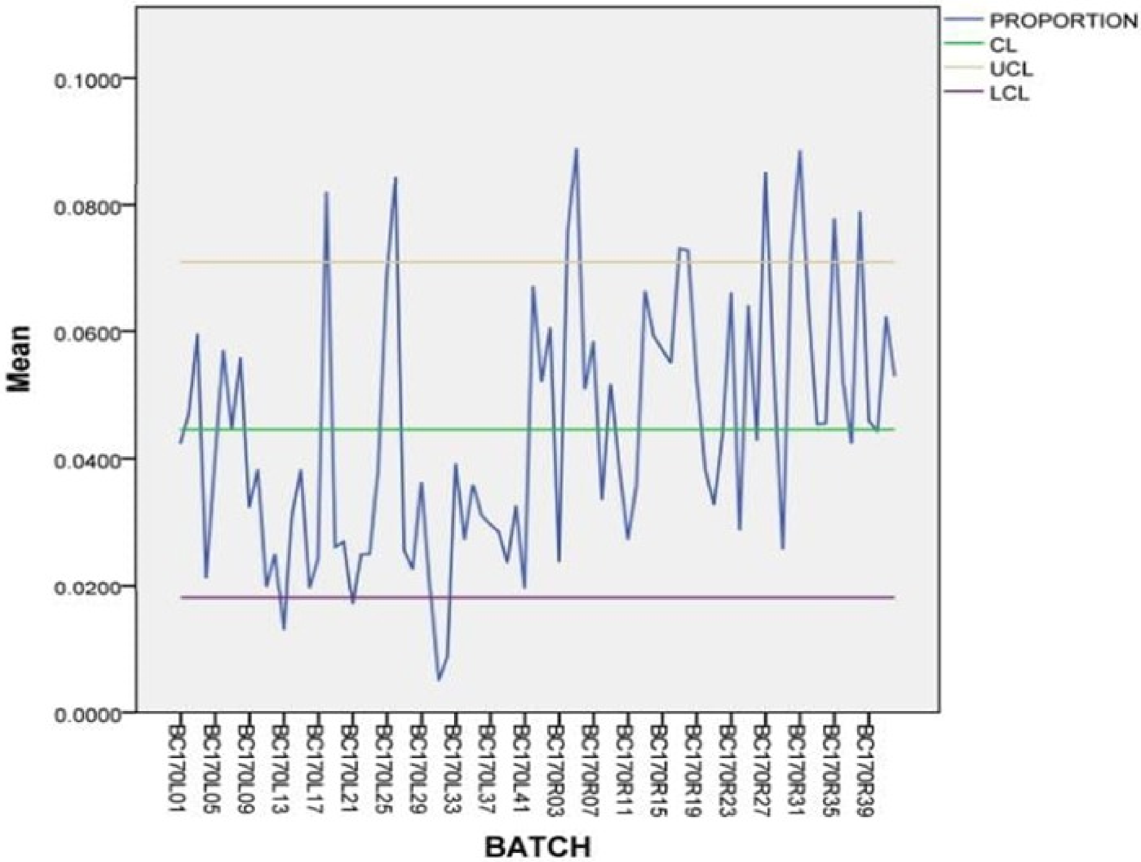

Figure 6.1: P Chart before SCADA implementation

actions and process improvement.

Overall, the P-chart highlights that the existing system is unstable and requires a real-time monitoring solution like SCADA to achieve process control, reduce variability, and improve product quality.

Chapter 7

PERFORMANCE ANALYSIS AFTER SCADA

After implementing the SCADA (Supervisory Control and Data Acquisition) system in the Electronic Testing Department (ETD) at HLL Lifecare Ltd., a significant improvement in process performance and quality control was observed. The transition from a manual monitoring system to a real-time automated system enhanced data accuracy, reduced delays, and improved overall efficiency.

7.1 IMPROVEMENT IN MONITORING AND CONTROL

The SCADA system enabled real-time monitoring of all ETD units, allowing operators and supervisors to continuously track critical parameters such as voltage, current, and machine status. This eliminated the dependence on manual observation and ensured immediate detection of deviations.

Continuous data collection and centralized monitoring improved visibility across all machines, making it easier to identify performance issues and take corrective actions promptly.

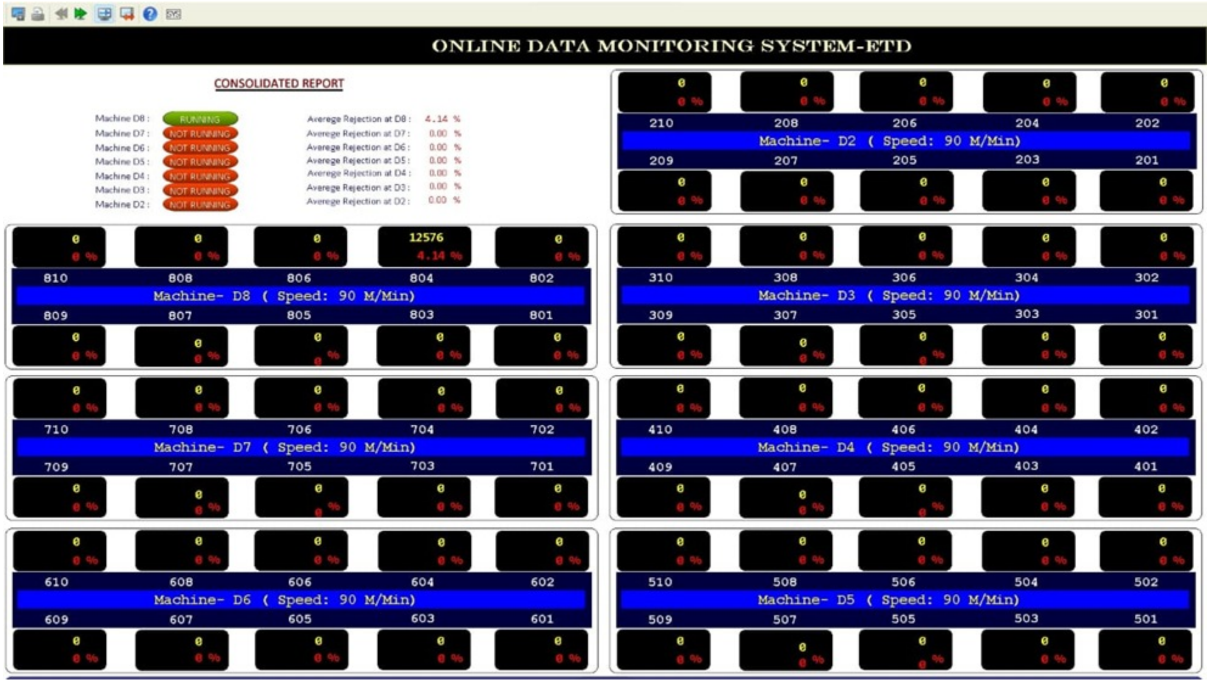

Figure 7.1: Real-Time Machine Performance Dashboard

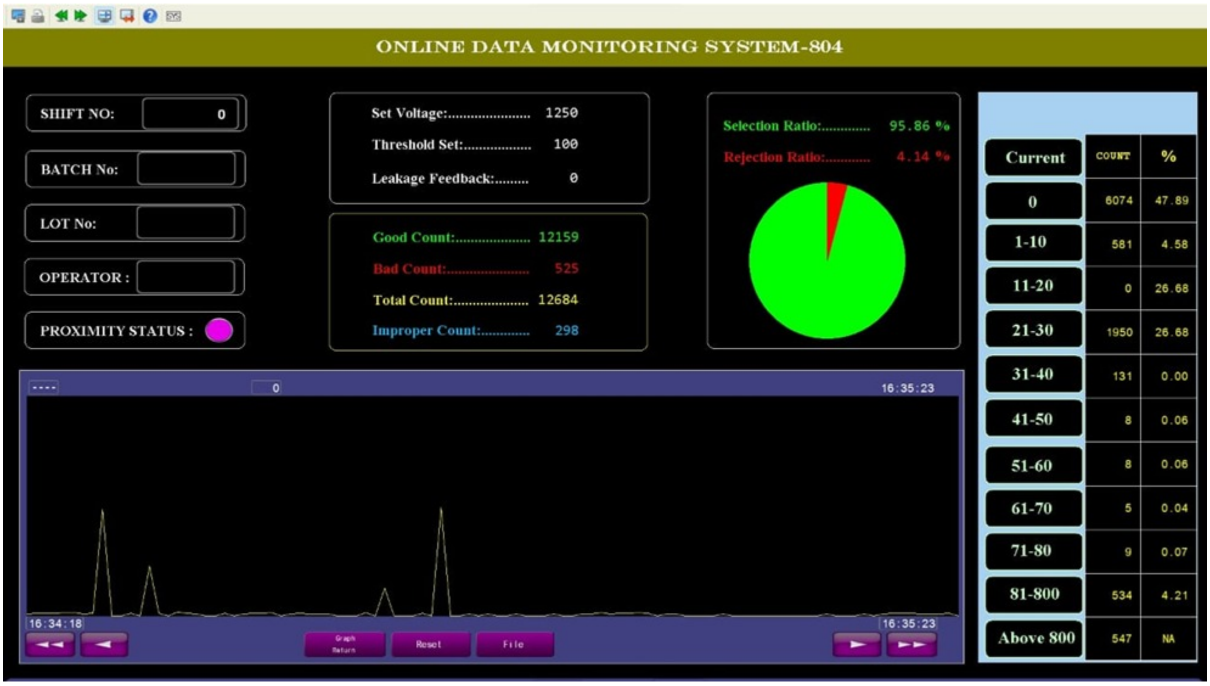

Figure 7.2: Online Data Monitoring System

7.2 ENHANCEMENT IN DATA MANAGEMENT

With SCADA implementation, data collection became fully automated and centralized. Parameters such as production count, rejection rate, and machine performance are now recorded accurately and stored for analysis.

This improvement increased:

In-depth analysis capability from approximately 50% to higher levels Real-time data monitoring from around 40% to nearly 100Centralized data collection from 20% to complete integration

The availability of accurate and real-time data supports better decision-making and process optimization.

7.3 IMPROVEMENT IN DECISION MAKING

SCADA provides instant access to process data through dashboards and visualizations, enabling faster and more accurate decision-making. Operators can respond immediately to alarms and warnings, reducing delays in identifying issues.

Decision-making efficiency improved significantly, as the system provides clear insights into process performance and deviations.

7.4 REDUCTION IN DELAY AND REJECTION RATE

One of the major improvements observed after SCADA implementation is the reduction in delay in identifying issues. Previously, delays were as high as 80%, but with real-time alerts and monitoring, this has been significantly reduced.Additionally, the rejection rate decreased from approximately 10% to around 2%, indicating improved accuracy in product classification and reduced false rejection of good-quality condoms.

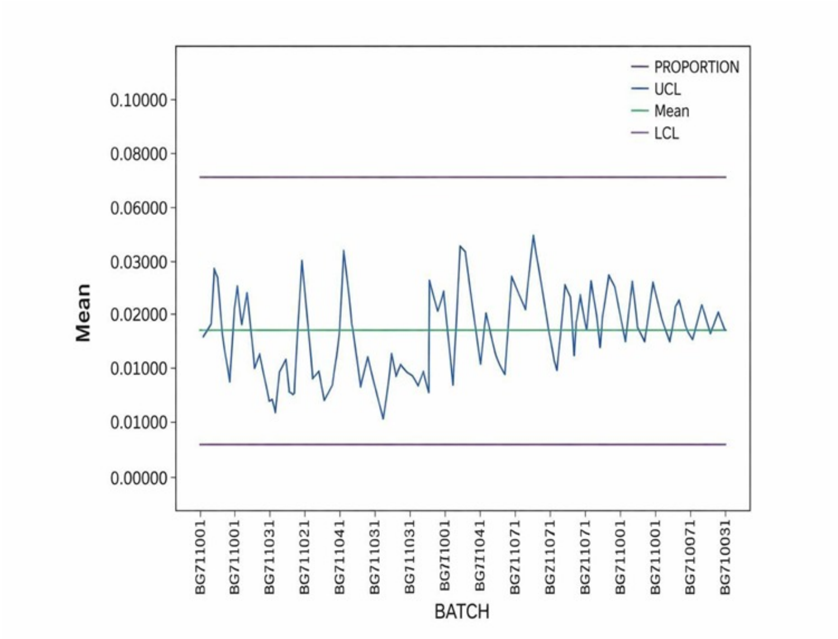

Figure 7.3: P Chart - post SCADA implentation

7.5 STATISTICAL ANALYSIS USING P-CHART

To evaluate the effectiveness of the SCADA system, a P-chart was constructed after implementation to analyze the proportion of defective products across different batches.

The P-chart clearly shows that all data points fall within the Upper Control Limit (UCL) and Lower Control Limit (LCL), indicating that the process is statistically stable and under control. Unlike the pre-SCADA condition, where multiple points exceeded control limits, the post-SCADA chart demonstrates improved process consistency.

The variation in defect proportion has significantly reduced, and the data points are closely clustered around the mean value. This indicates that the process performance has become more consistent and predictable.

Additionally, there are no abnormal patterns such as trends, cycles, or sudden spikes in the chart. This suggests that only random (common cause) variation is present, and no special causes are affecting the process.

Overall, the P-chart confirms that the implementation of SCADA has successfully stabilized the process and improved quality control.

7.6 IMPACT OF INTRODUCING SCADA

The introduction of the SCADA system has resulted in significant improvements across various performance parameters in the Electronic Testing Department (ETD). Key performance indicators show remarkable enhancement, with in-depth analysis improving from approximately 50% to 80%, real-time data monitoring increasing from around 40% to nearly 100%, and decision-making efficiency rising from 40% to about 90%. Centralized data collection has improved from 20% to full integration, while the delay in identifying issues has been reduced significantly from 80% to around 30%. Additionally, the rejection rate has decreased from 10% to approximately 2%, and overall productivity has increased from 60% to nearly 100%, clearly demonstrating the effectiveness of SCADA in enhancing operational performance.

Furthermore, the system has contributed to a reduction in process variability and defects through real-time monitoring and automated data collection, ensuring consistent application of critical parameters such as voltage and current. This has improved classification accuracy and minimized false rejection of good-quality products, leading to reduced material waste and cost savings. SCADA has also transformed the process from a manual and reactive system into an automated and proactive one, enabling operators to make faster and more accurate decisions with immediate corrective actions. The significant reduction in delays ensures that issues are identified and resolved promptly, thereby improving overall process efficiency.

In addition, the centralized SCADA dashboard provides enhanced data visibility and control by allowing real-time monitoring of all ETD units. Graphical representations such as trend graphs and performance charts help in better understanding process behavior and identifying areas for improvement. Overall, the implementation of SCADA has led to improved process stability, reduced variability, increased productivity, better quality control, and enhanced decision-making capability. These improvements clearly demonstrate that SCADA is an effective solution for modernizing the quality control process in HLL Lifecare Ltd.

In depth analysis	~50%	~80%
Real time data monitoring	~40%	~100 %
Decision making	~40%	~90%
Centralised data collection	~20%	~100 %
Delay in identifying issues	~80%	~30%
Rejection rate	~10%	~2%
Productivity	~60%	~100 %

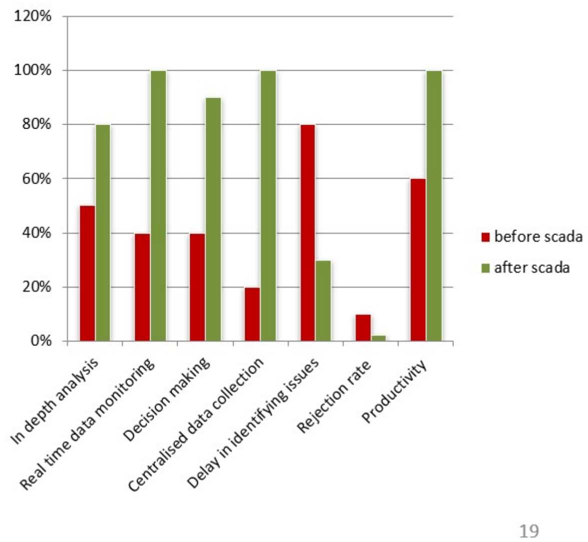

Figure 7.4: Before vs. After Performance Comparison in ETD

Chapter 8

RESULT

The implementation of a SCADA-based real-time quality monitoring system at HLL is designed to transform the Electronic Testing Department (ETD) and overall production process. By integrating advanced sensors, data acquisition, and automation, the system is expected to deliver tangible improvements in quality, efficiency, and resource utilization. The expected outcomes are elaborated below:

8.1 IMPLEMENTATION OF SCADA ENABLED REAL-TIME MONITORING OF ETD UNITS

The implementation of the SCADA system has enabled real-time monitoring of all Electronic Testing Department (ETD) units, marking a significant shift from traditional manual monitoring practices. Previously, operators had to rely on periodic checks and manual recording of data, which often led to delays and inaccuracies. With SCADA, all process parameters are continuously tracked and displayed in real time through a centralized interface.

This real-time visibility allows operators and supervisors to instantly observe the operational status of each machine, including whether it is running, idle, or facing any fault. Parameters such as voltage and current are updated continuously, providing a clear picture of the testing conditions. This ensures that any abnormality can be identified at the exact moment it occurs.

Moreover, the system enhances coordination between different levels of operation. Supervisors can monitor multiple ETD units simultaneously without physically being present at each location. This reduces dependency on manual supervision and improves overall control of the production process.

Another important advantage is the reduction in human error. Since data collection and monitoring are automated, the chances of missing critical information or recording incorrect values are minimized. This leads to more reliable and accurate monitoring of the process.

Overall, real-time monitoring through SCADA has improved transparency, responsiveness, and efficiency in ETD operations. It enables a proactive approach to quality control, ensuring that issues are addressed immediately rather than after significant losses have occurred.

8.2 REDUCTION IN PROCESS VARIABILITY OBSERVED IN POST-IMPLEMENTATION P-CHART

The post-implementation P-chart clearly indicates a significant reduction in process variability after the introduction of SCADA. In the earlier system, variations in process parameters were frequent due to inconsistent monitoring and lack of control. These variations resulted in fluctuations in product quality across different batches.

With SCADA, continuous monitoring ensures that critical parameters such as voltage and current are maintained within specified limits. This reduces fluctuations caused by machine inconsistencies and environmental factors such as humidity and temperature.

The clustering of data points around the mean in the P-chart reflects improved consistency in the process. This means that the production process is operating under stable conditions, producing uniform quality products.

Additionally, reduced variability leads to better predictability of process outcomes. When variations are minimized, it becomes easier to maintain quality standards and meet production targets consistently.

Thus, the reduction in process variability is a direct outcome of improved monitoring, control, and timely corrective actions enabled by the SCADA system.

8.3 ALL SAMPLE POINTS FALL WITHIN CONTROL LIMITS (UCL LCL) → PROCESS STABILITY ACHIEVED

One of the most significant outcomes observed after SCADA implementation is that all sample points fall within the Upper Control Limit (UCL) and Lower Control Limit (LCL). This indicates that the process is statistically stable and under control.

In the pre-SCADA condition, several points exceeded control limits, indicating the presence of assignable causes such as machine errors or environmental variations. However, after implementing SCADA, such deviations have been eliminated.

Process stability is essential for ensuring consistent product quality. When all points lie within control limits, it means that only natural or random variations are present, and no abnormal conditions are affecting the process.

The ability to maintain process stability also reduces the need for frequent adjustments and rework. This improves efficiency and reduces production costs.

Therefore, achieving statistical control through SCADA is a major milestone in improving the reliability and effectiveness of the quality control system.

8.4 FALSE REJECTION OF GOOD-QUALITY PRODUCTS REDUCED SIGNIFICANTLY

False rejection of good-quality products was a major issue in the previous system, primarily due to inaccurate measurements and inconsistent monitoring. Variations in environmental conditions and sensor misalignment often led to incorrect classification of products.

With the implementation of SCADA, the monitoring of critical parameters has become more accurate and consistent. This ensures that products are classified correctly based on precise data rather than assumptions or manual interpretation.

The reduction in false rejection has a direct impact on material utilization. Fewer good-quality products are discarded, leading to reduced wastage and improved resource efficiency.

This improvement also contributes to cost savings, as less material is lost due to incorrect rejection. It enhances overall profitability and makes the production process more sustainable.

Thus, SCADA plays a crucial role in improving classification accuracy and reducing unnecessary losses in production.

8.5 IMPROVED DATA ACCURACY AND CENTRALIZED DATA STORAGE

SCADA significantly improves data accuracy by automating the data collection process. In the earlier system, data was recorded manually, which often resulted in errors, omissions, and inconsistencies.

With SCADA, all process data is captured directly from sensors and stored automatically in a centralized database. This eliminates human intervention and ensures that the data is accurate and reliable.

Centralized data storage also allows easy access to historical data for analysis and reporting. This helps in identifying trends, patterns, and recurring issues in the process.

The availability of accurate data supports better decision-making and continuous improvement initiatives. It enables management to evaluate performance and implement corrective measures effectively.

Overall, improved data accuracy and centralized storage enhance the efficiency and effectiveness of the quality control system.

8.6 FASTER FAULT DETECTION AND CORRECTIVE ACTION THROUGH ALARMS ALERTS

The SCADA system includes an advanced alarm and alert mechanism that plays a crucial role in fault detection. Whenever a parameter exceeds its predefined limit, the system immediately generates a warning or alarm.

This instant notification allows operators to quickly identify the source of the problem and take corrective action. In the earlier system, delays in detecting faults often resulted in increased defects and production losses.

Faster fault detection minimizes the impact of issues on the production process. Problems can be resolved before they escalate, ensuring smooth operation.

The system also reduces downtime by enabling quick troubleshooting and maintenance. This improves machine availability and overall productivity.

Thus, the alarm and alert system in SCADA enhances responsiveness and ensures efficient process control.

8.7 ENHANCED OPERATOR EFFICIENCY AND DECISION-MAKING CAPABILITY

The implementation of SCADA has significantly improved operator efficiency by simplifying monitoring and control tasks. Operators no longer need to manually check each machine, as all information is available on a centralized dashboard.

This reduces workload and allows operators to focus on critical tasks rather than routine monitoring. It also enables them to manage multiple machines simultaneously.

The availability of real-time data improves decision-making capability. Operators can quickly analyze the situation and take appropriate actions based on accurate information.

This leads to faster response times and better control over the process. It also reduces the chances of errors caused by delayed or incorrect decisions.

Overall, SCADA enhances operator performance and contributes to improved productivity.

8.8 OVERALL DEFECT PROPORTION REDUCED AND PROCESS CONSISTENCY IMPROVED

The overall defect proportion has reduced significantly after the implementation of SCADA. This indicates an improvement in product quality and process control.

Consistent monitoring and control of process parameters ensure that defects are minimized. The system maintains stable operating conditions, leading to uniform product quality.

Improved process consistency also enhances customer satisfaction, as products meet quality standards consistently. This is especially important in healthcare manufacturing.

The reduction in defects also reduces rework and waste, improving efficiency and cost-effectiveness.

Thus, SCADA contributes to achieving a stable, reliable, and high-quality production process.

Chapter 9

CONCLUSION

The implementation of a SCADA-based monitoring system in the Electronic Testing Department (ETD) at HLL Lifecare Ltd. has brought a significant transformation in the quality control process. By introducing real-time monitoring of critical parameters such as voltage, current, and machine status, the system has enhanced the visibility and control of the production process. This real-time capability ensures that any deviation from standard conditions is immediately identified, allowing for prompt corrective actions and minimizing the chances of defective products reaching the next stage.

One of the most important outcomes of this project is the successful transition from a manual and semi-automated monitoring system to a fully automated and integrated system. In the earlier setup, manual observation and recording of data often led to errors, delays, and inconsistencies. With the adoption of SCADA, these limitations have been effectively eliminated. The system ensures accurate data collection, reduces dependency on human intervention, and enhances the reliability of the entire process.

The role of real-time data acquisition in improving process efficiency cannot be overstated. Continuous monitoring enables early detection of abnormalities such as fluctuations in voltage or current, which are critical indicators of product defects. By identifying such issues at an early stage, the system helps in minimizing material waste and reducing false rejection of good-quality products. This not only improves resource utilization but also contributes to significant cost savings for the organization.

Another key achievement of the SCADA implementation is the stabilization of the production process. Statistical analysis using control charts clearly shows that the process has shifted from an out-of-control state to a statistically stable condition. All process variations are now within control limits, indicating that the system is operating under controlled conditions with minimal variability. This stability is essential for maintaining consistent product quality, especially in healthcare manufacturing where standards are stringent.

The system has also contributed to enhanced operator efficiency and improved decision-making capabilities. With access to real-time data and graphical dashboards, operators can quickly analyze process conditions and take appropriate actions. This reduces response time and improves the overall effectiveness of process management. Additionally, centralized data storage allows for better analysis, reporting, and continuous improvement initiatives.

Furthermore, the SCADA system provides a scalable and flexible solution that can be extended to other departments and industries. Its ability to integrate with existing systems, monitor multiple parameters, and provide actionable insights makes it a valuable tool for modern manufacturing environments. The approach adopted in this project can serve as a model for other quality-sensitive industries such as pharmaceuticals, food processing, and medical device manufacturing.

In conclusion, the SCADA-based monitoring system has successfully addressed the major challenges in the ETD process, including false rejection, process variability, and lack of real-time monitoring. It has improved quality control, increased productivity, reduced costs, and ensured consistent product quality. The project demonstrates the importance of adopting advanced automation technologies in achieving operational excellence and aligns with the principles of Industry 4.0, paving the way for smarter and more efficient manufacturing systems.

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